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(54) **SWITCHGEAR GROUND AND TEST DEVICE HAVING FIRST AND SECOND SETS OF GROUNDING BARS AND GROUND SHOE ASSEMBLY**

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H02B 1/16 (2006.01)

(52) **U.S. Cl.**
CPC **G01R 31/3272** (2013.01); **H02B 1/16** (2013.01)

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CPC G01R 31/3272; H02B 1/16
See application file for complete search history.

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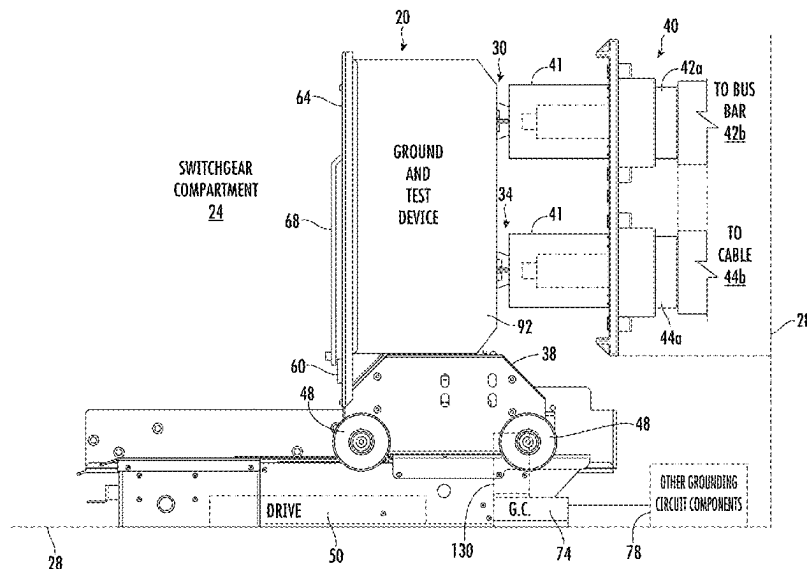
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(57) **ABSTRACT**

A ground and test (G&T) device includes a test device housing having upper terminals and lower terminals carried by the test device housing and engaging the load and line conductors when the test device housing is installed within the compartment of the switchgear frame. A plurality of grounding bars include a first set of grounding bars that connect the upper terminals to a lower ground bar and a second set of grounding bars that connect the lower terminals to the lower ground bus bar when the first set of grounding bars are not connected. A ground shoe assembly is connected to the lower ground bus bar and configured to engage a grounding circuit carried by the switchgear frame and includes a ground shoe bracket, bus bars, and die springs.

17 Claims, 15 Drawing Sheets



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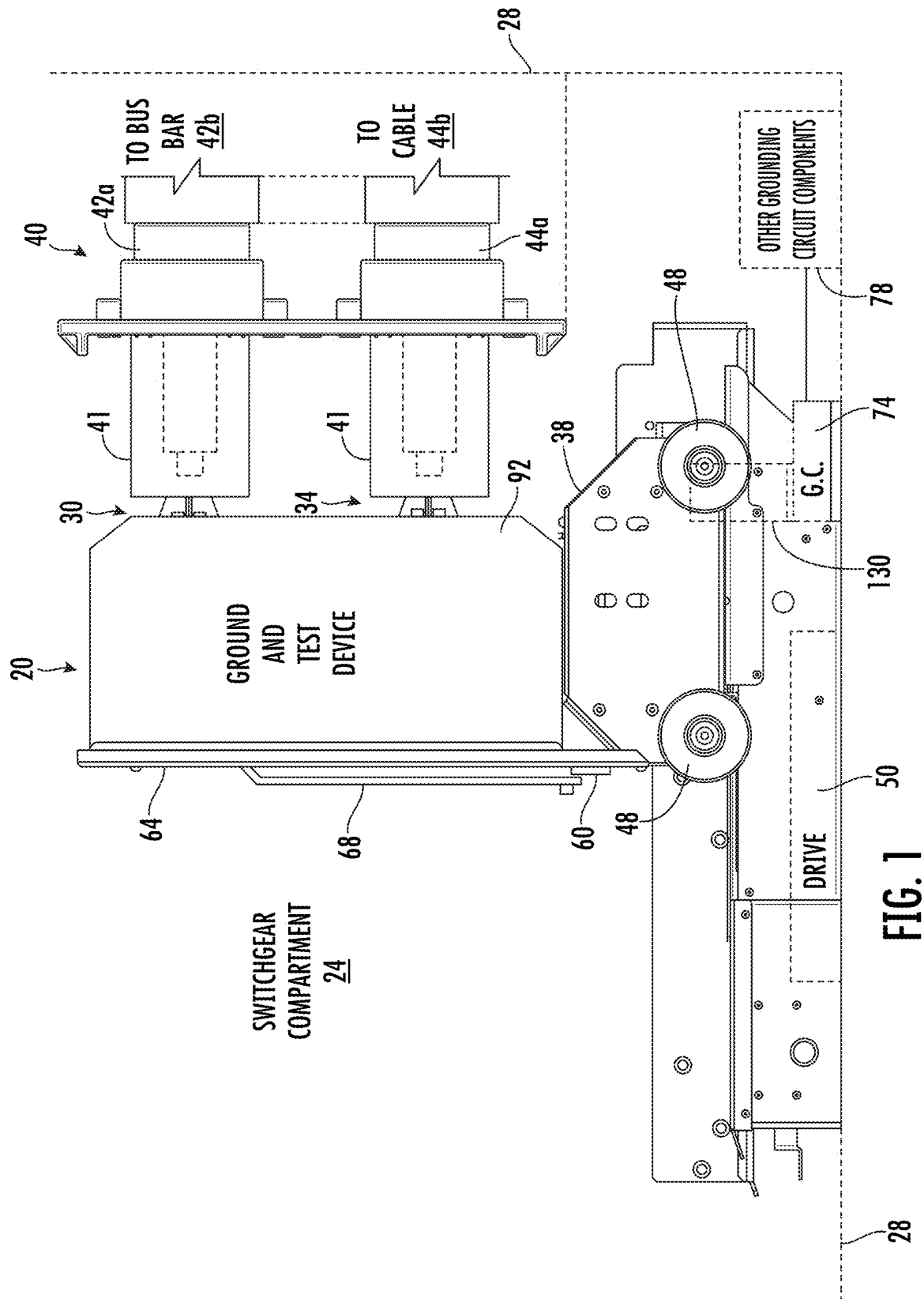
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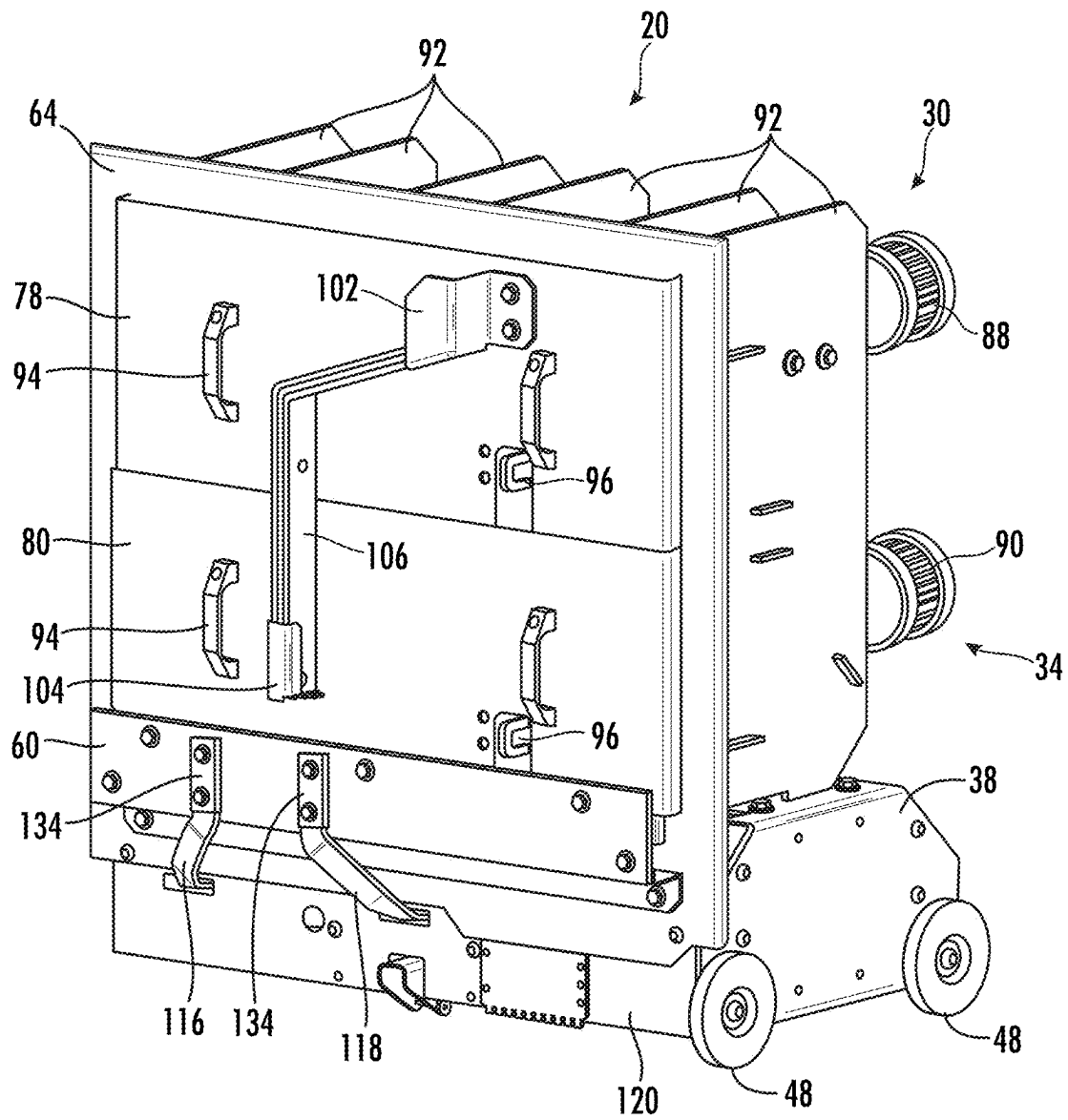
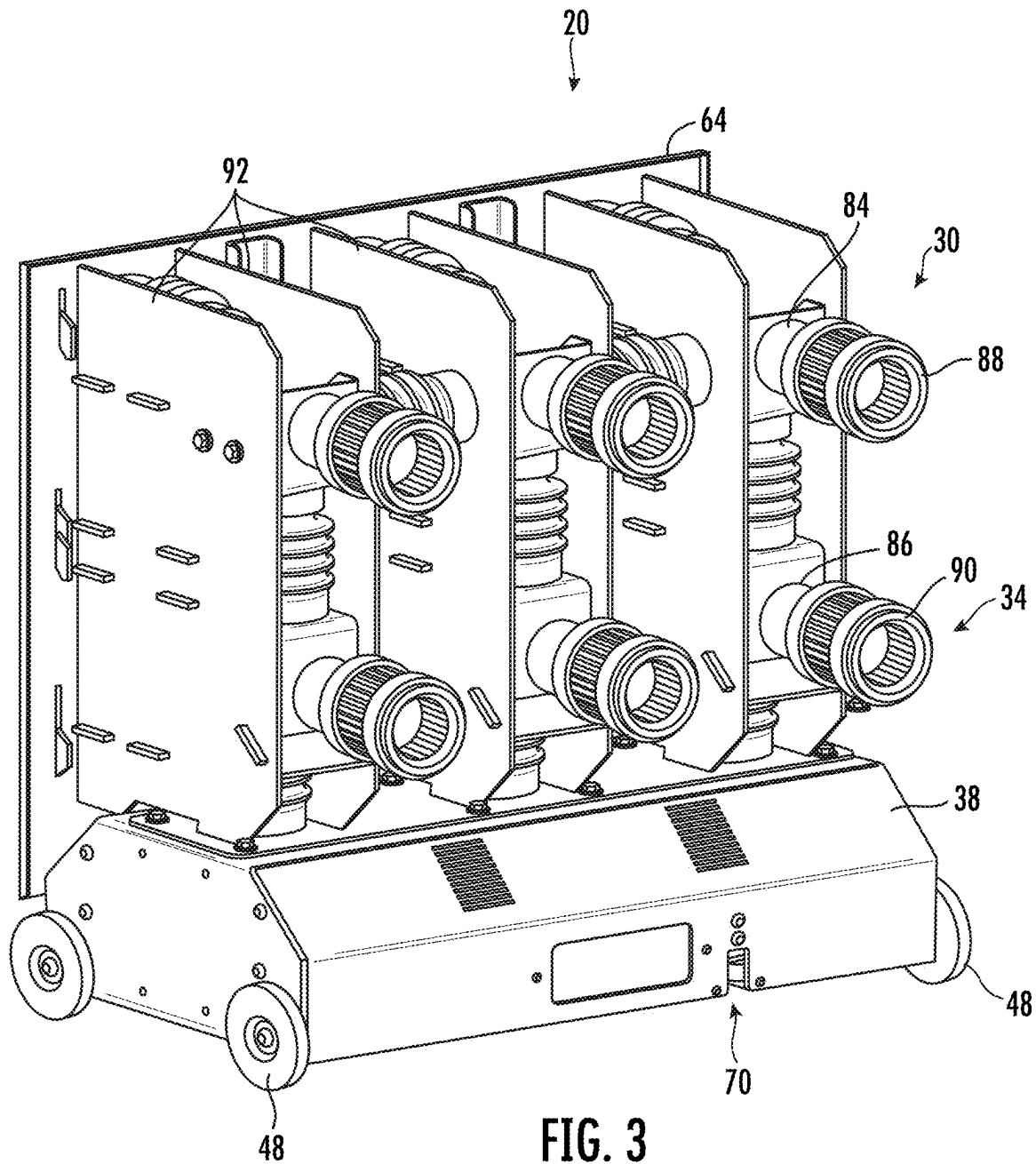


FIG. 2



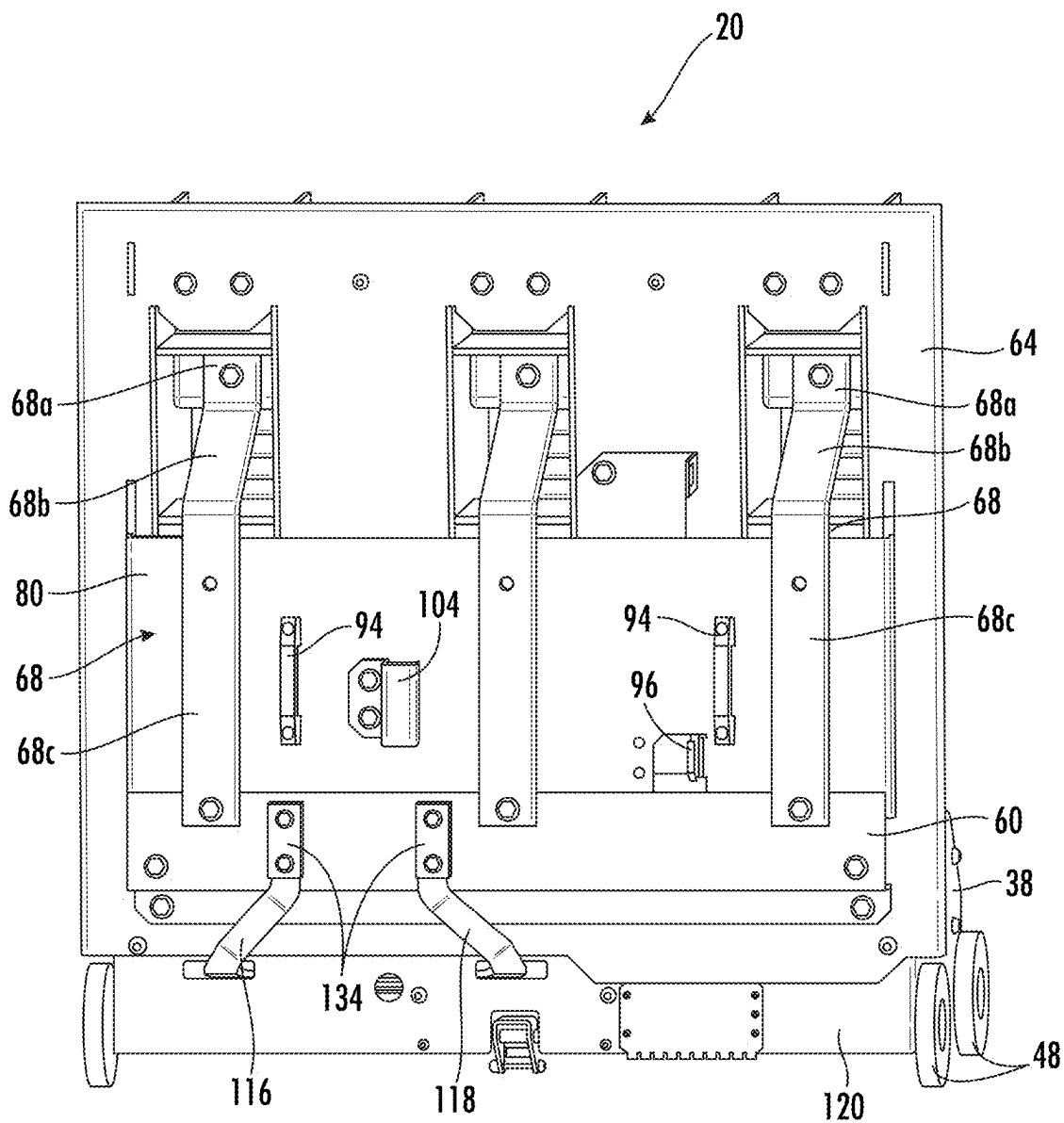


FIG. 4

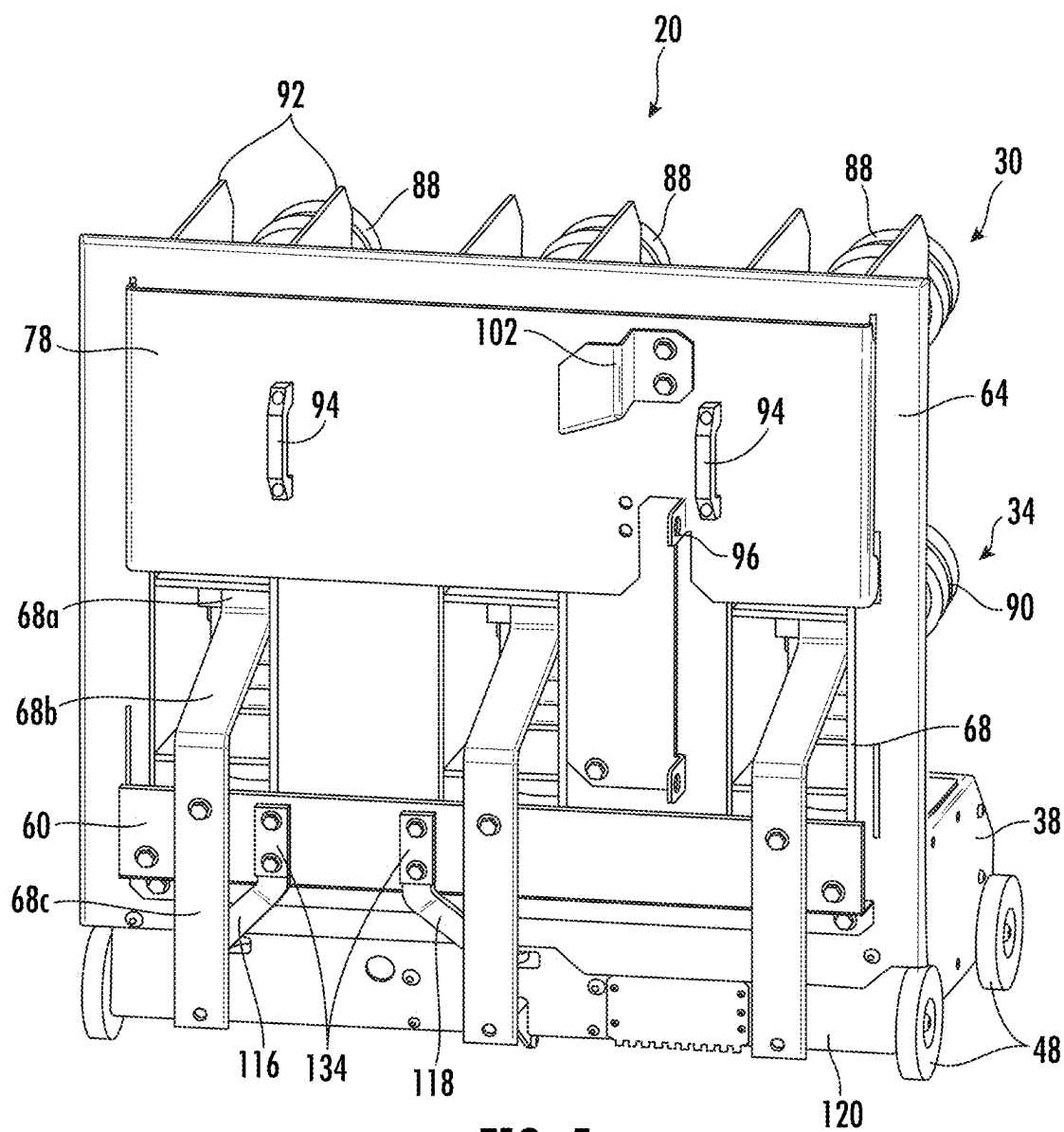


FIG. 5

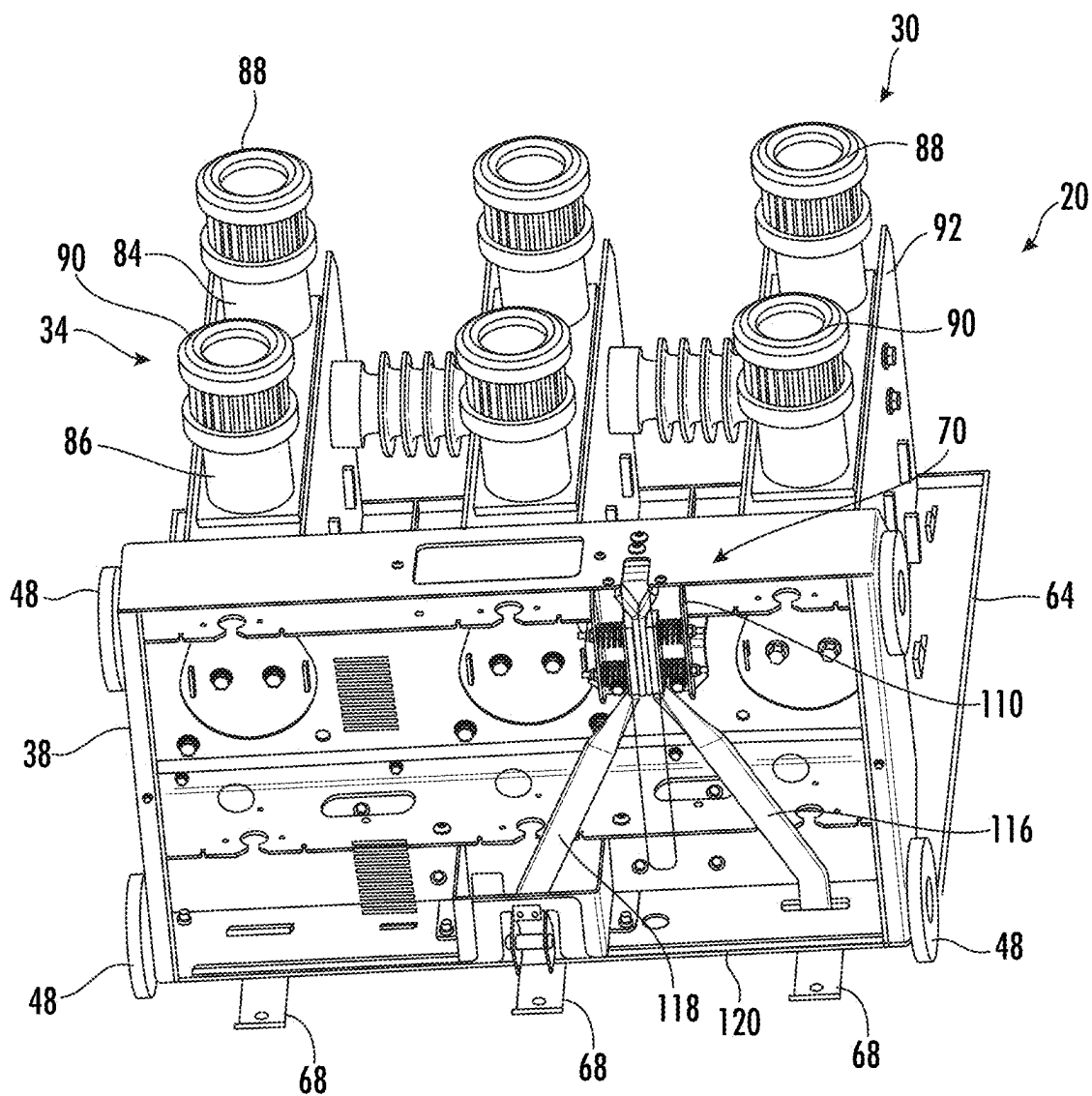
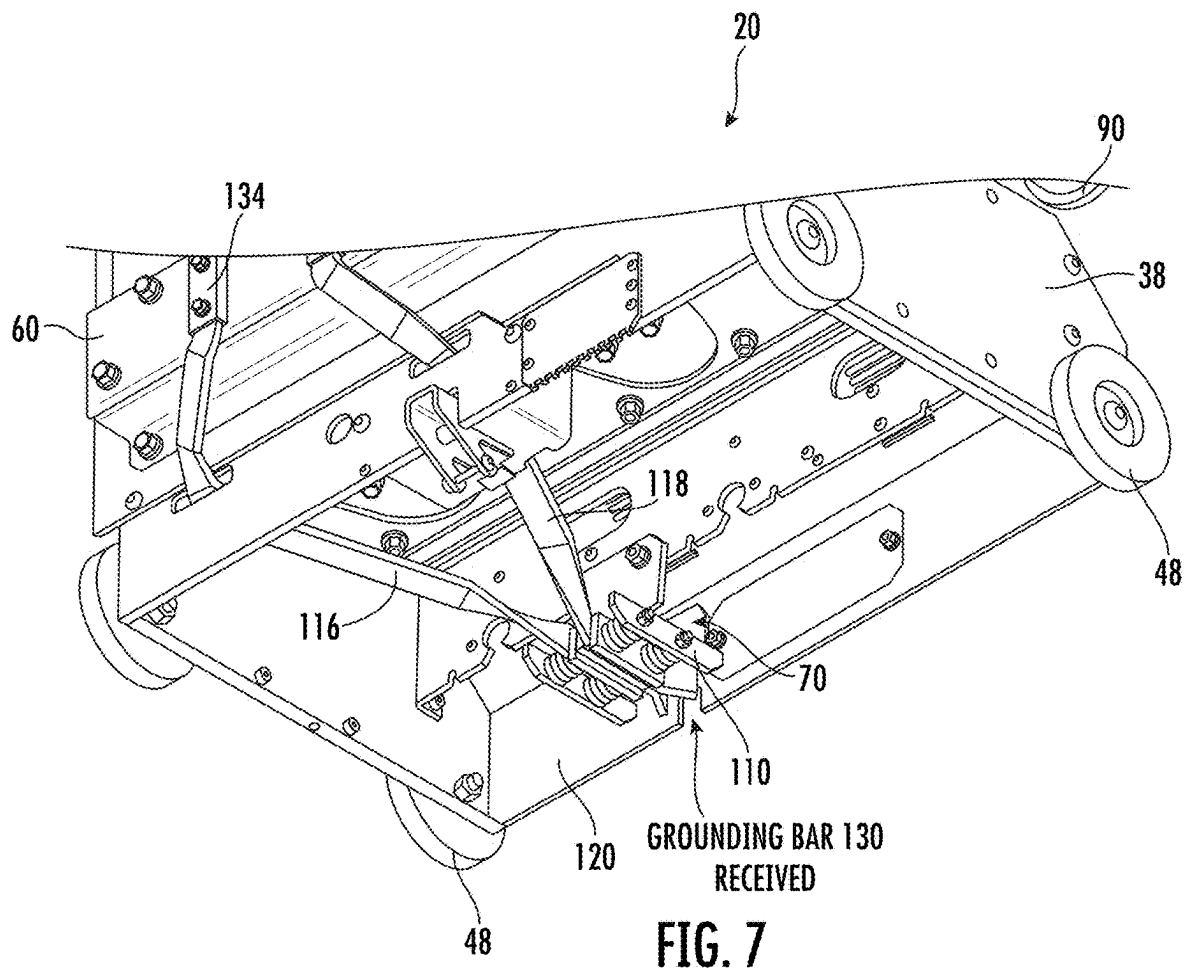


FIG. 6



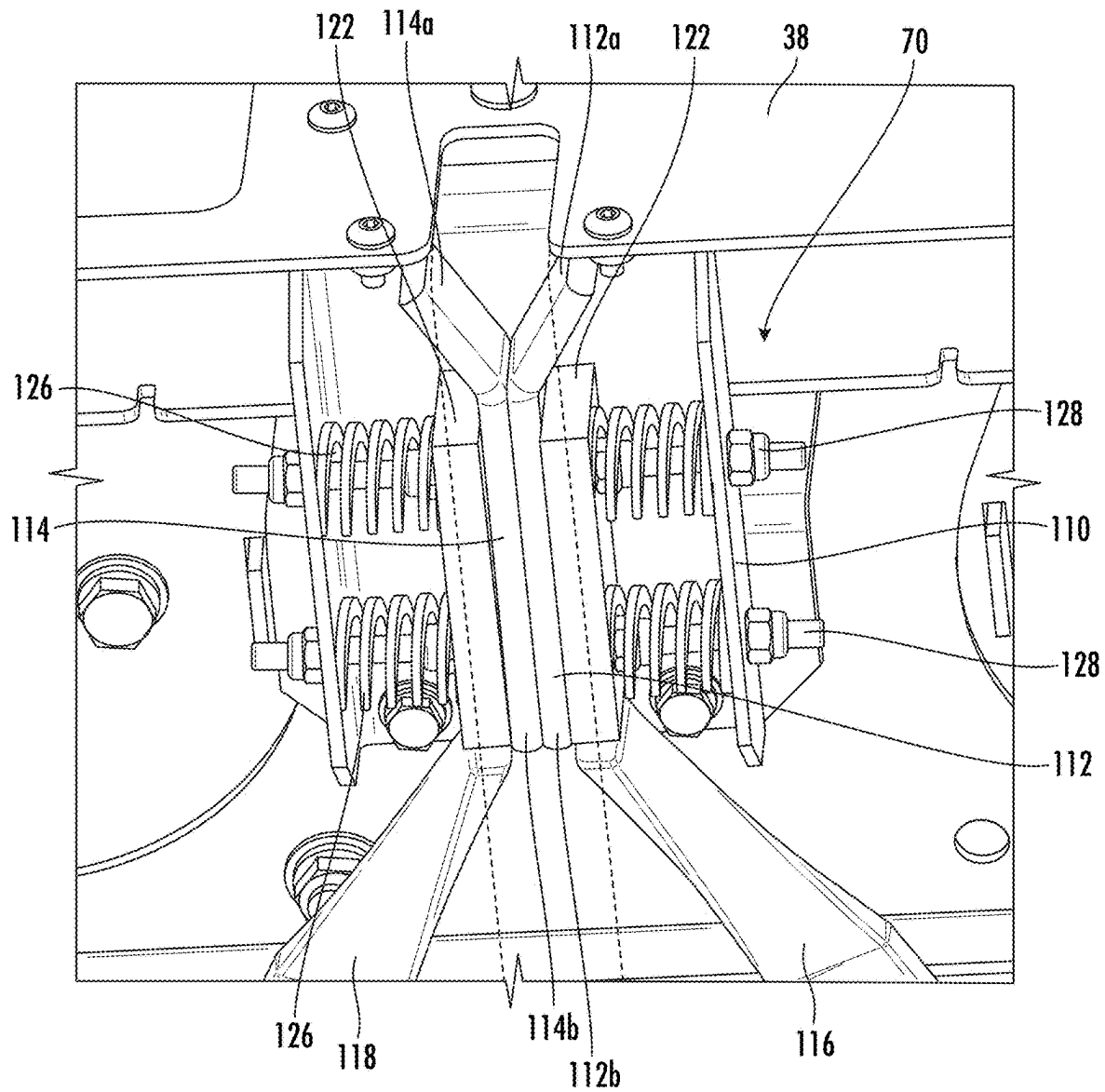


FIG. 8

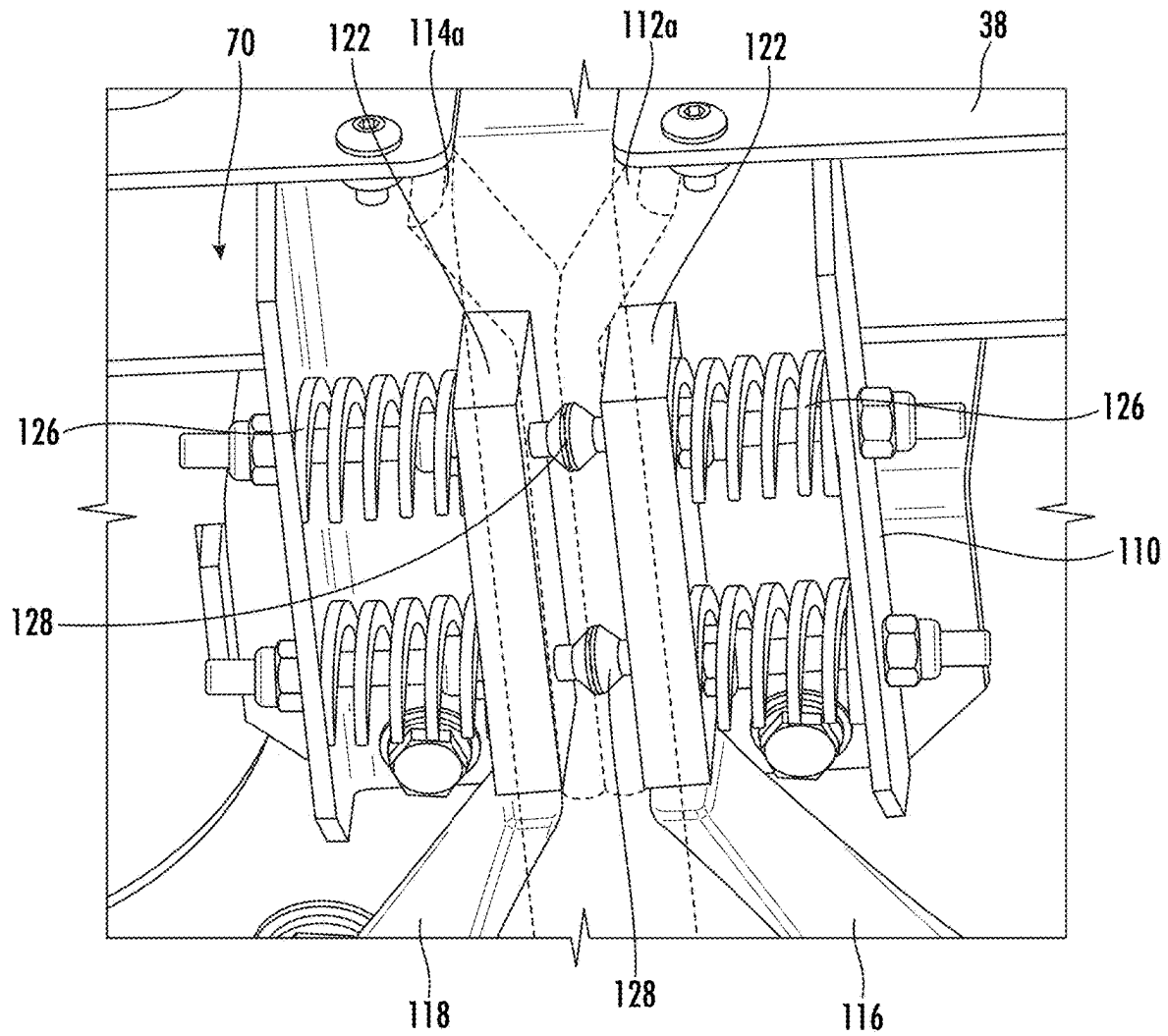


FIG. 9

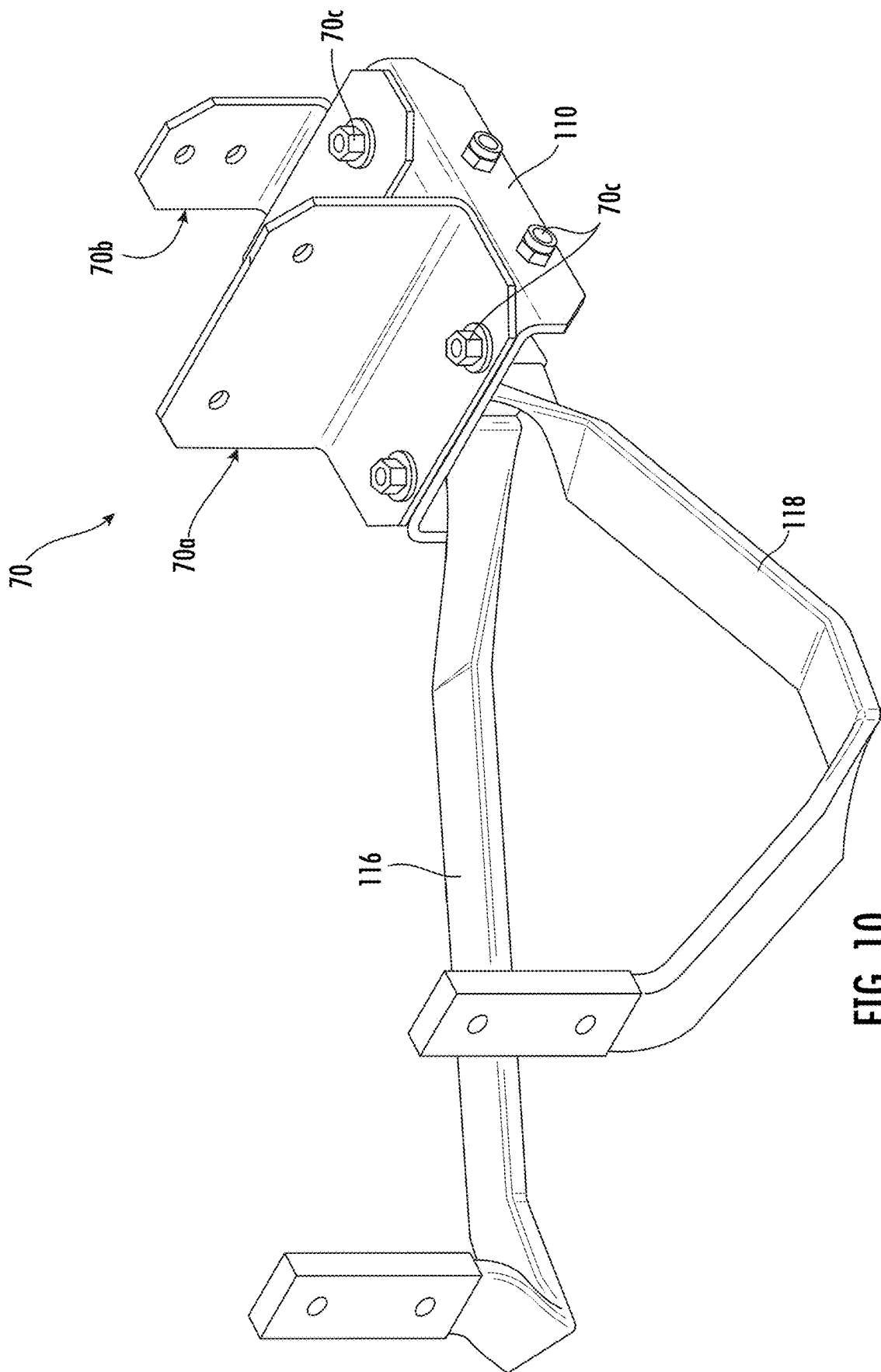
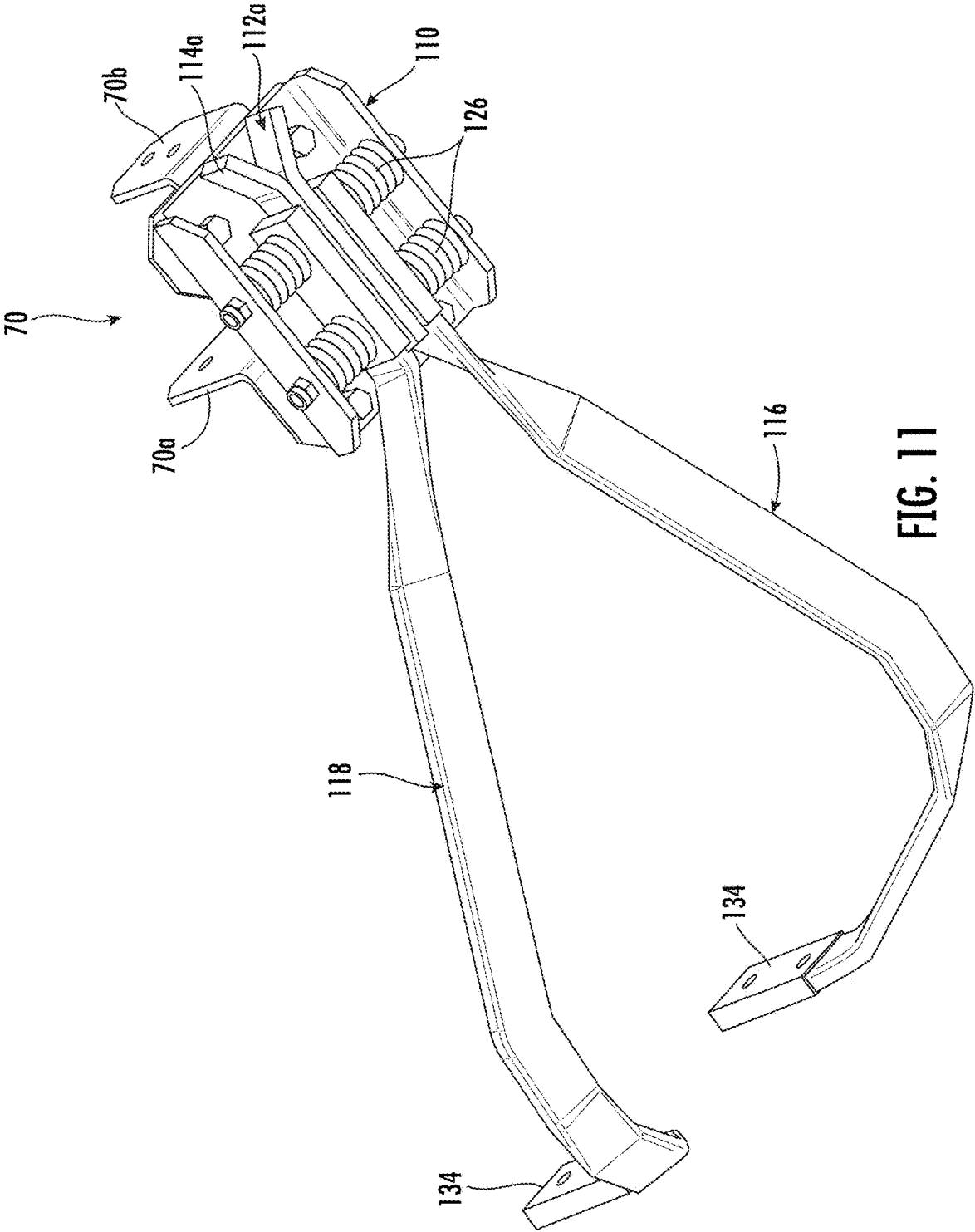


FIG. 10



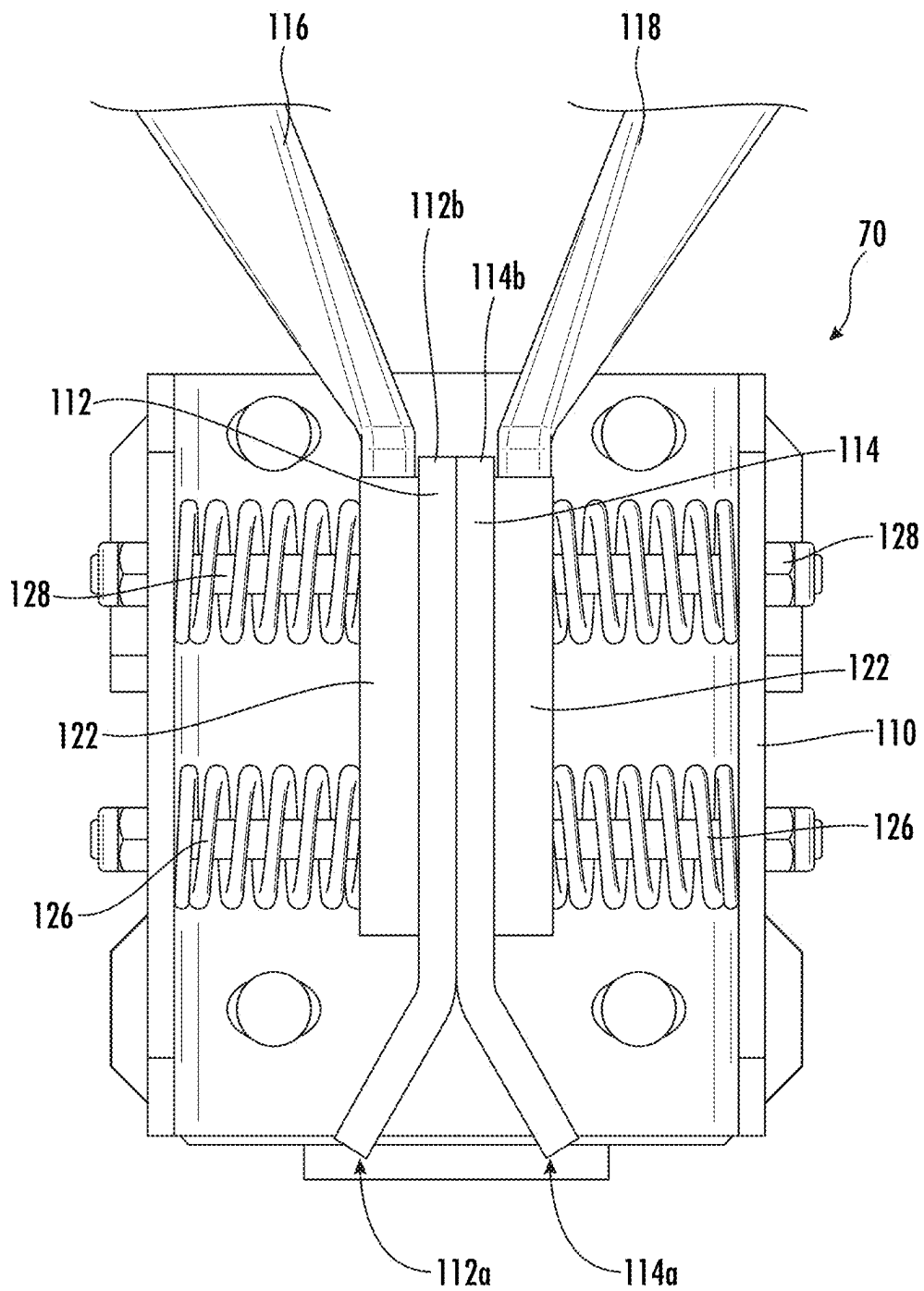


FIG. 12

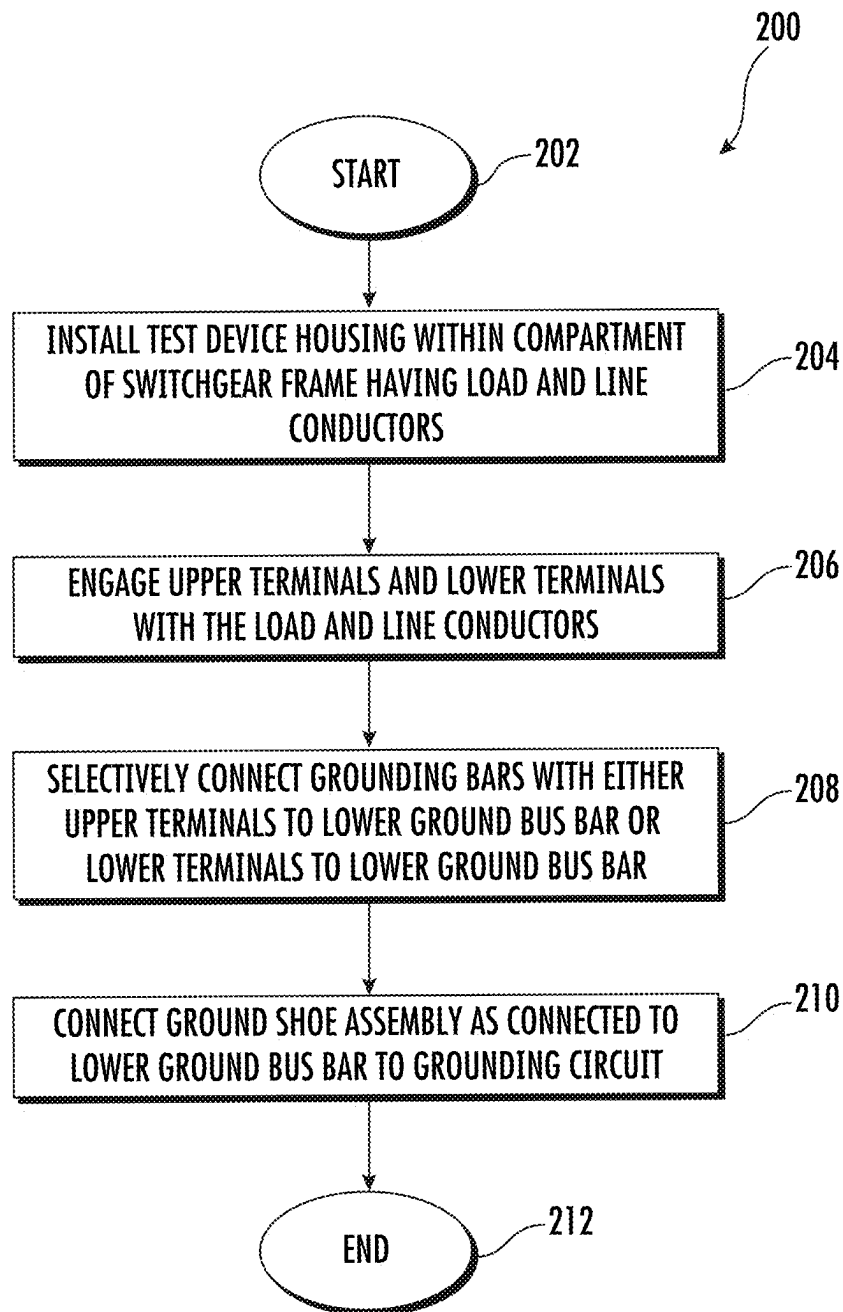


FIG. 13

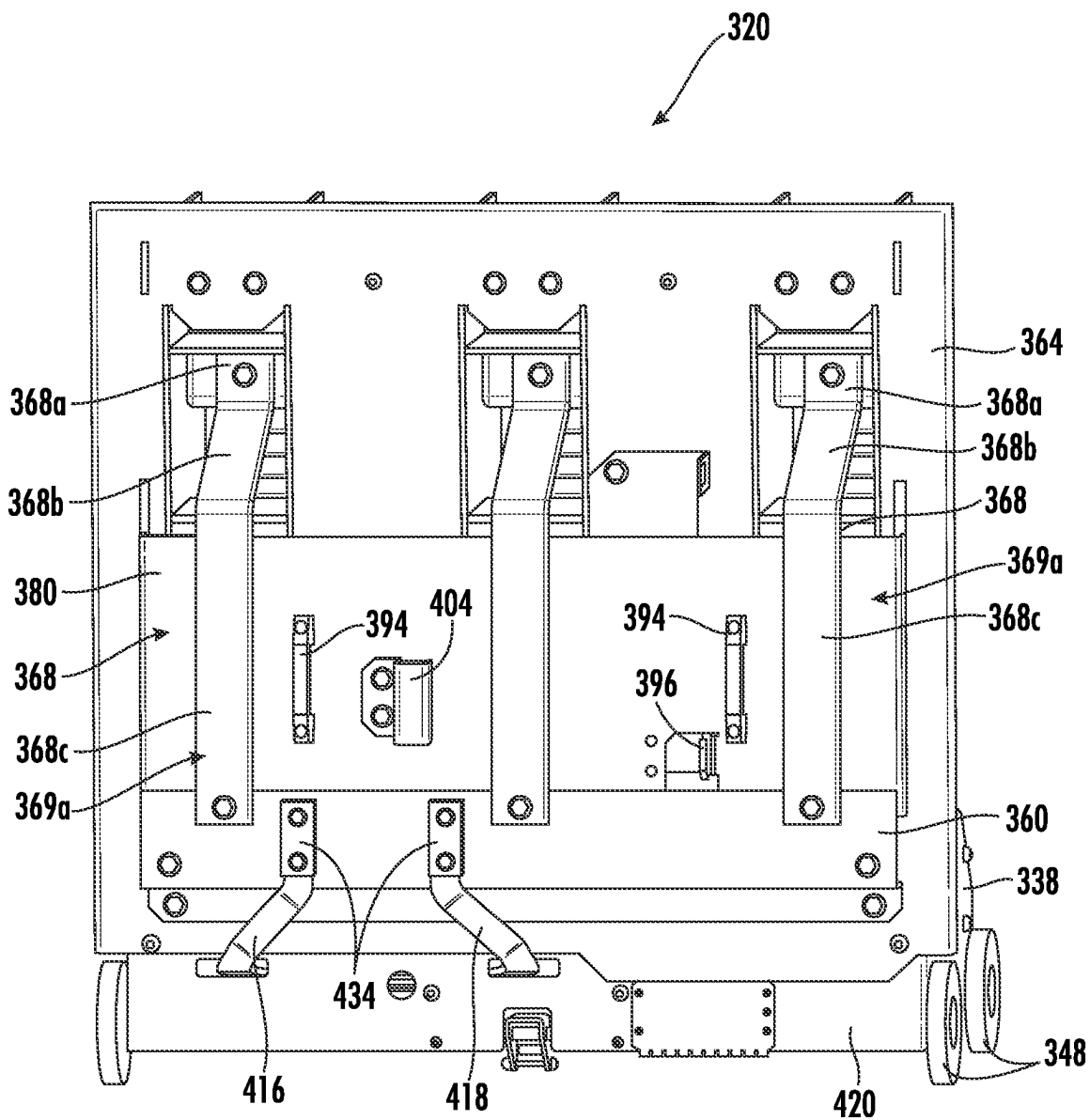


FIG. 14

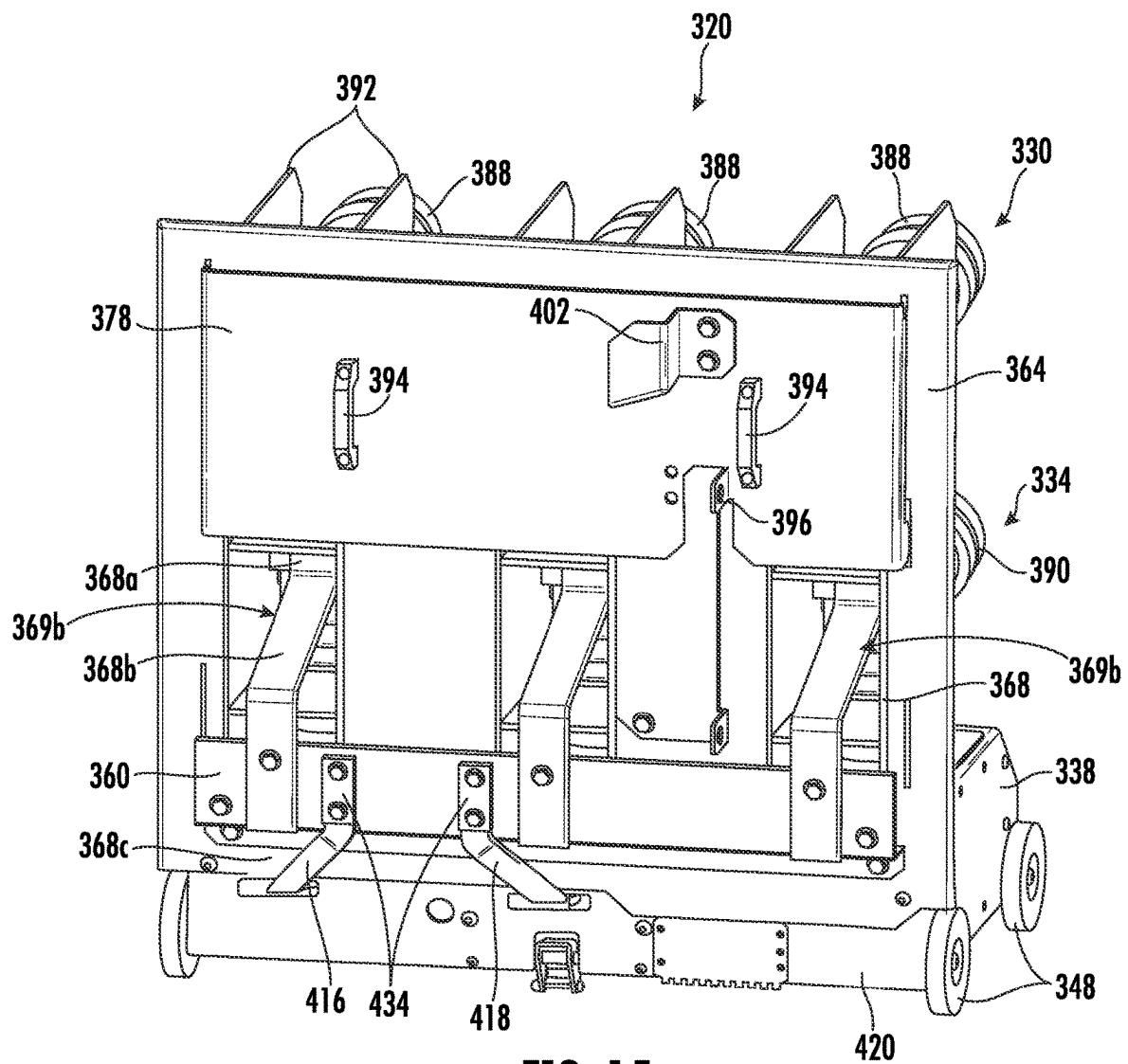


FIG. 15

1

SWITCHGEAR GROUND AND TEST DEVICE HAVING FIRST AND SECOND SETS OF GROUNDING BARS AND GROUND SHOE ASSEMBLY

PRIORITY APPLICATION(S)

This is a continuation-in-part (CIP) application based upon U.S. patent application Ser. No. 18/366,724 filed Aug. 8, 2023, which is a continuation application based upon U.S. patent application Ser. No. 17/651,069 filed Feb. 15, 2022 (now U.S. Pat. No. 11,762,020), the disclosures which are hereby incorporated by reference in their entirety.

FIELD OF THE INVENTION

The present invention relates to the field of electronic systems, and more particularly, this invention relates to ground and test (G&T) devices used with switchgear systems.

BACKGROUND OF THE INVENTION

Metal-clad or metal-enclosed medium voltage switchgear systems operate as three-phase systems that connect to the three-phase power distribution grid and provide various control functions and provide protection against short circuit events and similar overcurrent or other fault conditions. They often include circuit breakers, which open and close individual circuits and are mounted on a truck that is movable within a compartment of a switchgear frame. Circuit breakers may include vacuum interrupters, with a respective vacuum interrupter forming a single-phase circuit breaker for one of the three-phases.

The vacuum interrupters are removable from the switchgear frame by disengaging the circuit breaker from the load and line conductors and racking out the truck carrying the vacuum interrupters from the switchgear compartment. The metal-clad switchgear is compartmentalized and uses grounded metal components for many parts of the circuit to help minimize fault propagation and to make repairs easier and safer. An electrical bus may be insulated to minimize a potential flash over or fault between phases of circuits within the switchgear system.

During maintenance, the conventional and standard electrical safety practices require the insertion of a temporary grounding device within the switchgear compartment to protect the electrician from accidental energization of power equipment in the system and permit grounding of any high voltages that may remain on a line. It is necessary to ground the high voltage electric lines to protect personnel working on these lines and protect circuit components. It is also desirable to be able to test phase out, voltages, and selectively ground either the bus or line circuits during maintenance and testing.

There are numerous designs for ground and test devices used in switchgear systems. These designs include: 1) simple manual; 2) complex manual; 3) simple electrical; and 4) complex electrical. A simple manual ground and test device usually includes upper and lower terminals that connect load and line conductors to a ground circuit. Some simple manual ground and test devices use flexible cables that knock each other during operation of the ground and test device and strip insulation from the cables, creating a safety hazard. Also, some grounding mechanisms employed in

2

these ground and test devices do not provide an adequate ground connection when in use.

SUMMARY OF THE INVENTION

This summary is provided to introduce a selection of concepts that are further described below in the Detailed Description. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

In general, a ground and test (G&T) device may comprise a test device housing configured to be removably installed within a compartment of a switchgear frame having load and line conductors. A plurality of upper terminals and plurality of lower terminals may be carried by the test device housing and configured to engage the load and line conductors when the test device housing is installed within the compartment of the switchgear frame. A lower ground bus bar may be carried by the test device housing. A plurality of grounding bars may selectively connect either the upper terminals to the lower ground bus bar or connect the lower terminals to the lower ground bus bar. The plurality of grounding bars include a first set of grounding bars that connect the upper terminals to the lower ground bus bar when the first set of grounding bars are not connected.

A ground shoe assembly may be connected to the lower ground bus bar and configured to engage a grounding circuit carried by the switchgear frame. The ground shoe assembly includes a ground shoe bracket and first and second opposing bus bars each having first and second ends and carried by the ground shoe bracket. The first ends engage the grounding circuit. First and second flexible cables interconnect each of the second ends to the lower ground bus bar. Die springs are carried by the ground shoe bracket and exert a biasing force against first and second bus bars.

The G&T device may comprise three upper terminals and three lower terminals configured to engage respective three-phase load and line conductors. Each grounding bar in first and second sets may comprise an upper flat end that is configured to engage the respective upper or lower terminal, an outwardly angled section that extends outward from the test device housing, and a vertically extending lower flat section that is configured to engage the lower ground bus bar. The test device housing may include a front panel having removable upper and lower doors to expose upper and lower terminals onto which the grounding bars are secured. The upper and lower terminals may comprise upper and lower contact arms each carrying cluster assemblies that engage load and line conductors.

Each of the first and second opposing bus bars include flared ends that extend outward, and the grounding circuit includes a grounding bar over which the flared first ends engage as the test device housing is inserted within the compartment of the switchgear frame. The first and second flexible cables each may comprise a braided cable.

A method aspect of operating a ground and test (G&T) device may comprise installing a test device housing within a compartment of a switchgear frame having load and line conductors. The method may include engaging a plurality of upper terminals and a plurality of lower terminals carried by the test device housing with the load and line conductors, and selectively connecting a plurality of grounding bars with either the upper terminals to a lower ground bus bar carried by the test device housing or connecting the lower terminals to the lower ground bus bar. The grounding bars comprise a first set of grounding bars that connect the upper terminals

3

to the lower ground bus bar and a second set of grounding bars that connect the lower terminals to the lower ground bus bar when the first set of grounding bars are not connected. The method includes connecting a ground shoe assembly to the lower ground bus bar and to a grounding circuit carried by the switchgear frame. The ground shoe assembly includes a ground shoe bracket and first and second opposing bus bars each having first and second ends and carried by the ground shoe bracket, the first ends engaging the grounding circuit. First and second flexible cables interconnect each of the second ends to the lower ground bus bar. Die springs are carried by the ground shoe bracket and exert a biasing force against first and second bus bars.

BRIEF DESCRIPTION OF THE DRAWINGS

Other objects, features and advantages of the present invention will become apparent from the Detailed Description of the invention which follows, when considered in light of the accompanying drawings in which:

FIG. 1 is a side elevation view of the ground and test device showing upper and lower terminals connected to the load and line conductors in accordance with a non-limiting example.

FIG. 2 is a front isometric view of the ground and test device of FIG. 1.

FIG. 3 is a rear isometric view of the ground and test device of FIG. 1 showing the upper and lower contact arms each carrying arm and cluster assemblies.

FIG. 4 is another front isometric view of the ground and test device showing the upper door removed and the grounding bars connected between the upper terminals and lower ground bus bar.

FIG. 5 is another front isometric view similar to FIG. 4, but showing the lower door removed and the grounding bars connected between the lower terminals and lower ground bus bar.

FIG. 6 is a bottom isometric view of the ground and test device showing the ground shoe assembly.

FIG. 7 is another bottom isometric view similar to FIG. 6, but showing a different view of the ground shoe assembly.

FIG. 8 is an enlarged bottom isometric view of the ground shoe assembly of FIGS. 6 and 7.

FIG. 9 is another enlarged bottom isometric view of the ground shoe assembly showing in transparency the first and second opposing bus bars and their connection with the ground shoe bracket.

FIG. 10 is an enlarged top isometric view of the ground shoe assembly.

FIG. 11 is an enlarged bottom isometric view of the ground shoe assembly.

FIG. 12 is an enlarged bottom plan view of the ground shoe assembly.

FIG. 13 is a high-level flowchart of an example method of operating the ground and test device.

FIG. 14 is a front isometric view of another embodiment of the ground and test device showing a first set of grounding bars that connect upper terminals to the lower ground bus bar.

FIG. 15 is a front isometric view similar to FIG. 14 showing a second set of grounding bars that connect lower terminals to the lower ground bus bar.

DETAILED DESCRIPTION

Different embodiments will now be described more fully hereinafter with reference to the accompanying drawings, in

4

which preferred embodiments are shown. Many different forms can be set forth and described embodiments should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope to those skilled in the art.

Referring now to FIG. 1, there is illustrated the ground and test (G&T) device 20 that is removably installed within a switchgear compartment 24 of a switchgear frame shown by the dashed line at 28. The G&T device 20 includes a plurality of upper terminals 30 and a plurality of lower terminals 34 carried by a test device housing 38 and configured to engage load and line conductors 40 in the switchgear compartment 24 of the switchgear frame 28 when the test device housing 38 is installed in the switchgear compartment 24 of the switchgear frame 28.

The test device housing 38 may be configured similar to a conventional truck carrying conventional vacuum interrupters and insertable within the switchgear component 24, thus allowing the test device housing to be racked in and racked out, whether via a drive mechanism or manually. The test device housing 38 includes in this example three upper terminals 30 and three lower terminals 34 and is configured to engage the respective three-phase load and line conductors 40. For example, as shown in FIG. 1, the individual line conductors 42a may connect to a bus bar 42b and individual load conductors 44a may connect to a cable 44b. The upper terminals 30 may engage the line conductors 42a and connect to the bus bar 42b, which connect to outside high voltage sources, while the lower terminals 34 may connect to the load conductors 44a, and in turn, may connect to the cable 44b that connects to an apartment building or other load as non-limiting examples.

The test device housing 38 includes wheels 48 that allow the test device housing to be inserted within the switchgear compartment 24 of the switchgear frame 28. The wheels may roll on side rails (not shown) within the switchgear compartment 24 of the switchgear frame 28 and allow full insertion of the test device housing 38 such that the upper and lower terminals 30,34 engage the load and line conductors 40.

Although not illustrated in detail, the switchgear frame 28 supports a drive mechanism illustrated schematically by the dashed lines at 50 for each truck carrying a vacuum interrupter, and in turn, may be used to rack in and rack out a test device housing 38. Those skilled in the art should understand that the test device housing 38 may be racked in and racked out manually, such as by use of a hand crank or other mechanism or automatically via a drive mechanism 50. The test device housing 38 may have a width, height and depth that are similar in dimensions and configuration to a conventional truck carrying vacuum interrupters so that the test device housing may be received within the switchgear compartment 24 of the switchgear frame 28 with little difficulty.

Referring now more particularly to FIGS. 2-7, a lower ground bus bar 60 is carried by the test device housing 38 and formed in this example as a rectangular piece of bus that extends over the lower section of a front panel 64. A plurality of grounding bars 68 selectively connect either the upper terminals 30 to the lower ground bus bar 60 or connect the lower terminals 34 to the lower ground bus bar 60. A ground shoe assembly 70 is carried on the underside of the ground and test device 20 (FIGS. 6 and 7) and connected to the lower ground bus bar 60 and configured to engage a grounding circuit 74 (G.C.) carried by the switchgear frame 28. The grounding circuit 74 may connect to various switchgear and

5

other grounding components **78** to aid in grounding the switchgear frame **28** during maintenance and protect workers downstream from the switchgear frame, such as those working on downstream line components, e.g., a transformer or breaker.

In this example, the ground and test device **20** includes three upper terminals **30** and three lower terminals **34** configured to engage the respective three-phase load and line conductors **40**. Each grounding bar **68** is similarly configured and interchangeable with each other as best shown in example of FIGS. **4** and **5**. The same grounding bars **68** are used to engage the upper terminals **30** and connect to the lower ground bus bar **60** (FIG. **4**) or connect the lower terminals **34** to the lower ground bus bar (FIG. **5**). Each grounding bar **68** includes an upper flat end **68a** that is configured to engage either an upper terminal **30** or lower terminal **34**, and an outwardly angled section **68b** that extends outward from the test device housing **38**, and a vertically extending lower flat section **68c** that is configured at its lower end to engage the lower ground bus bar **60**.

The test device housing **38** includes its front panel **64** that may have a removable upper door **78** and removable lower door **80** to expose the respective upper and lower terminals **30,34** onto which the grounding bars **68** are secured. Upper terminals **30** are exposed when the upper doors **78** are removed and the lower terminals **34** are exposed when the lower doors **80** are removed. On the side opposite from which the grounding bars **68** connect, the upper and lower terminals **30,34** each include respective upper and lower contact arms **84,86**, which each carry respective upper and lower cluster assemblies **88,90** that engage the load and line conductors **40** (FIG. **3**).

The ground and test device **20** includes ground and test barrier panels **92** that separate the different upper and lower contact arms **84,86** and the respective upper and lower cluster assemblies **88,90** that engage the load and line conductors **40**. In this example, six vertical barrier panels **92** are employed (FIG. **3**). As illustrated, the test and device housing **38** and upper and lower cluster assemblies **88,90** and upper and lower contact arms **84,86** are similarly configured and dimensioned as a conventional truck carrying vacuum interrupters, such as disclosed in the incorporated by reference U.S. Pat. No. 11,735,893, such that when the ground and test device is inserted, the cluster assemblies **88,90** engage in a proper fit the load and line conductors, which include receptacles **41** that receive the cluster assemblies (FIG. **1**).

The test device housing **38** includes its front panel **64** having its removable upper and lower doors **78,80**, which each include pull handles **94** and a ground and test lock mechanism **96** to allow a padlock or locking mechanism to be inserted within the lock mechanism and prevent the locked doors **78,80** from being removed. The front panel **64** also includes a ground and test upper bus bracket **102** and lower bus bracket **104** and a ground and test grounding bus **106** connected between the two bus brackets.

Each grounding bar **68** may be formed as a single piece of metallic bus that is shaped in the configuration as shown in FIGS. **4** and **5** to include the upper flat end **68a**, the outwardly angled section **68b**, and the vertically extending lower flat section **68c**. Grounding bars **68** may vary in dimension, e.g., their length, width, and thickness, but in an example may range anywhere from 1/8 inch thick to 3/8 inch thick as found with bus metal used in switchgear systems and be about 1 inch to 3 inches in width. They may be long enough such that the grounding bars **68** may selectively connect the upper terminals **30** to the lower ground bus bar

6

60, but not extend well below the test device housing **38** to hit the floor of the switchgear frame **28** and interfere with movement of the test device housing when the grounding bars **68** are connected to the lower terminals **34** and to the lower ground bus bar as shown in FIG. **5**. The configuration where the grounding bars **68** are formed from a bus metal and similarly configured and interchangeable with each other is an advantage over those prior art ground and test devices that use a single flexible conductor to ground the upper or lower terminals.

Referring now to FIGS. **6-12**, further details of the ground shoe assembly **70** are illustrated. The ground shoe assembly **70** connects to the lower ground bus bar **60** and is configured to engage a grounding circuit **74** (FIG. **1**) carried by the switchgear frame **28** when the test device housing **38** is installed within the switchgear compartment **24** of the switchgear frame **28** and connected to load and line conductors **40**. Each ground shoe assembly **70** includes a ground shoe bracket **110** that is mounted on the underside of the test device housing **38** (FIGS. **6** and **7**). First and second opposing bus bars **112,114** each have first and second ends **112a, 112b** and **114a, 114b** (FIGS. **8** and **12**) and are carried by the ground shoe bracket **110**. The first ends **112a, 114a** engage the grounding circuit **74**. First and second flexible cables **116,118** interconnect each of the second ends **112b, 114b** of the first and second bus bars **112,114** to the lower ground bus bar **60** as best shown in FIGS. **6** and **7**, in which the first and second flexible cables extend through a blocking plate **120** positioned at the lower section of the front panel **64** and extend upward to connect onto the lower ground bus bar **60** (FIGS. **2-4**).

In an example, the first and second flexible cables **116,118** are formed as braided cable. As best shown in the enlarged bottom isometric views of the ground shoe assembly **70** in FIGS. **8, 9, 11**, and **12**, the flexible cables **116,118** formed as braided cable connect at their end to rectangular configured cable supports **122** that exert a biasing force against the opposing first and second opposing bus bars **112,114** through die springs **126** that are carried by the ground shoe bracket **110**, and thus, configured to bias first and second opposing bus bars against each other. As best shown in FIG. **9**, the bus bars **112,114** receive countersunk bolts **128** that retain the cable supports **122** and die springs **126** against the ground shoe bracket **110**. The first ends **112a, 114a** of each of the first and second opposing bus bars **112,114** include flared ends **112a, 114a** that extend outward. The grounding circuit **74** includes a vertical grounding bar **130** over which the flared ends **112a, 114a** engage as the test device housing **38** is inserted within the compartment of the switchgear frame **28** (FIGS. **1** and **7**).

A bottom cover plate (not shown) will cover the bottom section of the test device housing **38** such that the first and second flexible cables **116,118** and the ground shoe bracket **110** are not exposed. The cables **116,118** provide flexibility such that when the test device housing **38** is inserted within the compartment **24** of the switchgear frame **28**, the flared first ends **112a, 114a** engage the vertical grounding bar **130** and expand outward and separate the first and second opposing bus bars **112,114**. The four die springs **126** as illustrated bias the cable supports **122** and provide that sufficient contact exists between the grounding bar **130** and the first and second opposing bus bars **112,114** that connect onto the braided cables **116,118** and provide sufficient grounding surface contact.

As shown in FIG. **10**, the braided cables **116,118** as the flexible cables provide some rigidity if necessary, and include contact connectors **134** that attachment to the lower

ground bus bar **60** carried by the test device housing **38**. Other components of the ground shoe assembly **70** may include a ground shoe upper support plate **70a** and ground shoe model block bracket **70b** and fasteners **70c** (FIGS. **10** and **11**).

In operation, the ground and test device **20** may be inserted within the compartment **24** of a switchgear frame **28** that has the load and line conductors **40** and testing may occur first followed by grounding. The ground and test device **20** provides an additional measure of safety. The use of the grounding bars **68** that are similarly configured and interchangeable with each other facilitates maintenance and operator safety since there are no cables to be swapped. The same grounding bars **68** may be used for each of the different three-phase load and line conductors **40** to ground them via the grounding bars **68**.

Referring now to FIG. **13**, a high-level flowchart of an example method of operating the ground and test device **20** is illustrated at **200**. The process starts (Block **202**) and the test device housing **28** is installed within a switchgear compartment **24** of the switchgear frame **28** having load and line conductors **40** (Block **204**). A plurality of upper terminals **30** and a plurality of lower terminals **34** are carried by the test device housing **38** and engage the load and line conductors **40** (Block **206**). The method includes selectively connecting a plurality of grounding bars **68** with either the upper terminals **30** to the lower ground bus bar **60** carried by the test device housing **38** or connecting the lower terminals **34** to the lower ground bus bar (Block **208**). A ground shoe assembly **70** is connected to the lower ground bus bar **60** and engages the grounding **74** circuit carried by the switchgear frame **28** (Block **210**). The process ends (Block **212**).

The ground and test device **20** may be used for testing of target electrical apparatus for different test states. Both line and load circuits may be tested. The ground and test device **20** may be used to test utility, commercial, industrial, and residential environments, including distribution stations, main stations, power plants, shops and offices, commercial buildings, industrial production facilities, and similar units. Line voltage can be sensed.

Referring now to FIGS. **14** and **15**, there are illustrated a second embodiment of the ground and test device **320** that has first and second sets of grounding bars. Reference numbers are similar as in FIGS. **1-13**, except in the **300** and **400** series. The device **320** includes a plurality of grounding bars **368** that selectively connect either the upper terminals **330** to the lower ground bus bar **360** or connect the lower terminals **334** to the lower ground bus bar **360**. The plurality of grounding bars **368** includes a first set **369a** of grounding bars that connect the upper terminals **330** to the lower ground bus bar **360** and a second set **369b** of grounding bars that connect the lower terminals **334** to the lower ground bus bar when the first set of grounding bars are not connected. The ground shoe assembly **70** as described before is the same in configuration as described with reference to FIGS. **1-13**.

In an example, the first set **369a** of grounding bars **368** are similarly configured to the second set **369b** of grounding bars except the second set of grounding bars have a shorter, vertically extending lower flat section **368c**. There is also no central fastener opening for receiving a fastener as shown in FIGS. **4** and **5**, where the fastener opening is shown above where the grounding bars **68** are connected to the lower ground bus bar **60**. This configuration shown in FIGS. **14** and **15** provides a technical benefit since the current does not get necked down in the area by the central fastener opening

and begin overheating in that area. With the configuration in FIGS. **14** and **15**, a larger lower ground bus bar **360** is not required.

The ground and test device **320** in FIGS. **14** and **15** is similar as described above with reference to FIGS. **1-13** and includes the test device housing **338** configured to be removably installed within a compartment of a switchgear frame having load and line conductors. A plurality of upper terminals **330** and plurality of lower terminals **334** are carried by the test device housing **338** and configured to engage the load and line conductors when the test device housing is installed within a compartment of the switchgear frame. A lower ground bus bar **360** is carried by the test device housing **338**.

As noted before, the plurality of grounding bars **368** selectively connect either the upper terminals **330** to the lower ground bus bar **360** or connect the lower terminals **334** to the lower ground bus bar **360**. This plurality of grounding bars **368** includes the first set **369a** of grounding bars that connect the upper terminals **330** to the lower ground bus bar **360** and the second set **369b** of grounding bars that connect the lower terminals **334** to the lower ground bus bar when the first set of grounding bars are not connected. As noted before, the ground shoe assembly **70** is formed similar to that as shown in FIGS. **6-12** and as described above.

Each grounding bar **368** in first and second sets **369a,369b** includes an upper flat end **368a** that is configured to engage the respective upper or lower terminal **330,334**, and outwardly angled section **368b** that extends outward from the test device housing **338** and vertically extending lower flat section **368c** that is configured to engage the lower ground bus bar **360**. The test device housing **338** includes a front panel **364** having removable upper and lower doors **378,380** to expose upper and lower terminals **330,334**, which the grounding bars are secured. The upper and lower terminals **330,334** each include upper and lower contact arms, each carrying cluster assemblies **388,390** that engage load and line conductors. Other components are illustrated in FIGS. **14** and **15**, including the pull handle **394**, upper bus bracket **402**, lower bus bracket **404**, lock mechanism **396**, contact connectors **434**, first and second flexible cables **416,418**, blocking plate **420**, wheels **348**, and front panel **364**.

A method of operating the ground and test device **320** includes installing a test device housing **328** within a compartment of a switchgear frame having load and line conductors. The method includes engaging a plurality of upper terminals **330** and a plurality of lower terminals **334** carried by the test device housing **328** with the load and line conductors. The method further includes selectively connecting a plurality of grounding bars **368** with either the upper terminals **330** to the lower ground bus bar **360** or the lower terminals **334** to the lower ground bus bar. The plurality of grounding bars **368** includes a first set **369a** of grounding bars that connect the upper terminals **330** to the lower ground bus bar **360** and a second set **369b** of grounding bars that connect the lower terminals **334** to the lower ground bus bar when the first set of grounding bars are not connected. The method includes connecting a ground shoe assembly **70** (FIGS. **6-12**) to the lower ground bus bar **360** and configured to engage a grounding circuit carried by the switchgear frame.

As shown in FIGS. **6-12**, the ground shoe assembly includes a ground shoe bracket **110** and first and second opposing bus bars **112,114** each having first and second ends **112a,112b** and carried by the ground shoe bracket. The first ends **112a** engage the grounding circuit and first and second flexible cables **116,118** (**416** and **418** in FIGS. **14** and **15**)

interconnect each of the second ends to the lower ground bus bar. Die springs **126** are carried by the ground shoe bracket and may insert a biasing force against first and second bus bars **112,114** either upon ground and test device **320** insertion or only when insertion is completed.

Many modifications and other embodiments of the invention will come to the mind of one skilled in the art having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is understood that the invention is not to be limited to the specific embodiments disclosed, and that modifications and embodiments are intended to be included within the scope of the appended claims.

The invention claimed is:

1. A ground and test (G&T) device, comprising:
 - a test device housing configured to be removably installed within a compartment of a switchgear frame having load and line conductors;
 - a plurality of upper terminals and plurality of lower terminals carried by the test device housing and configured to engage the load and line conductors when the test device housing is installed within the compartment of the switchgear frame;
 - a lower ground bus bar carried by the test device housing;
 - a plurality of grounding bars that selectively connect either the upper terminals to the lower ground bus bar or connect the lower terminals to the lower ground bus bar, said plurality of grounding bars comprising a first set of grounding bars that connect the upper terminals to the lower ground bus bar and a second set of grounding bars that connect the lower terminals to the lower ground bus bar when the first set of grounding bars are not connected; and
 - a ground shoe assembly connected to the lower ground bus bar and configured to engage a grounding circuit carried by the switchgear frame, wherein said ground shoe assembly comprises a ground shoe bracket and first and second opposing bus bars each having first and second ends engaging the grounding circuit, and first and second flexible cables interconnecting each of said second ends to the lower ground bus bar, and die springs carried by said ground shoe bracket and exerting a biasing force against first and second bus bars.
2. The G&T device of claim 1, wherein each grounding bar in first and second sets comprises an upper flat end that is configured to engage the respective upper or lower terminal, an outwardly angled section that extends outward from the test device housing, and a vertically extending lower flat section that is configured to engage the lower ground bus bar.
3. The G&T device of claim 1, wherein said test device housing includes a front panel having removable upper and lower doors to expose upper and lower terminals onto which the grounding bars are secured.
4. The G&T device of claim 1, wherein said upper and lower terminals comprise upper and lower contact arms each carrying cluster assemblies that engage load and line conductors.
5. The G&T device of claim 1, wherein each of said first ends of each of said first and second opposing bus bars include flared ends that extend outward, and said grounding circuit includes a grounding bar over which the flared ends engage as the test device housing is inserted within the compartment of the switchgear frame.
6. The G&T device of claim 1, wherein said first and second flexible cables each comprises a braided cable.

7. A ground and test (G&T) device, comprising:
 - a test device housing configured to be removably installed within a compartment of a switchgear frame having three-phase load and line conductors, wherein said test device housing includes a front panel having removable upper and lower doors;
 - three upper terminals and three lower terminals carried by the test device housing and configured to engage the respective three-phase load and line conductors when the test device housing is installed within the compartment of the switchgear frame, each of said three upper terminals and three lower terminals configured to carry single-phase power, wherein the upper and lower terminals are configured such that upper terminals are exposed when the upper doors are removed and lower terminals are exposed when the lower doors are removed;
 - a lower ground bus bar carried by the test device housing;
 - a plurality of grounding bars that selectively connect either the upper terminals to the lower ground bus bar or connect the lower terminals to the lower ground bus bar, said plurality of grounding bars comprising a first set of grounding bars that connect the upper terminals to the lower ground bus bar and a second set of grounding bars that connect the lower terminals to the lower ground bus bar when the first set of grounding bars are not connected; and
 - a ground shoe assembly connected to the lower ground bus bar and configured to engage a grounding circuit carried by the switchgear frame, wherein said ground shoe assembly comprises a ground shoe bracket and first and second opposing bus bars each having first and second ends engaging the grounding circuit, and first and second flexible cables interconnecting each of said second ends to the lower ground bus bar, and die springs carried by said ground shoe bracket and exerting a biasing force against first and second bus bars.
8. The G&T device of claim 7, wherein each grounding bar in first and second sets comprises an upper flat end that is configured to engage the respective upper or lower terminal, an outwardly angled section that extends outward from the test device housing, and a vertically extending lower flat section that is configured to engage the lower ground bus bar.
9. The G&T device of claim 7, wherein said upper and lower terminals comprise upper and lower contact arms each carrying cluster assemblies that engage load and line conductors.
10. The G&T device of claim 7, wherein each of said first ends of each of said first and second opposing bus bars include flared ends that extend outward, and said grounding circuit includes a grounding bar over which the flared ends engage as the test device housing is inserted within the compartment of the switchgear frame.
11. The G&T device of claim 7, wherein said first and second flexible cables each comprises a braided cable.
12. A method of operating a ground and test (G&T) device, comprising:
 - installing a test device housing within a compartment of a switchgear frame having load and line conductors;
 - engaging a plurality of upper terminals and a plurality of lower terminals carried by the test device housing with the load and line conductors;
 - selectively connecting a plurality of grounding bars with either the upper terminals to the lower ground bus bar or the lower terminals to the lower ground bus bar, said

11

plurality of grounding bars comprising a first set of grounding bars that connect the upper terminals to the lower ground bus bar and a second set of grounding bars that connect the lower terminals to the lower ground bus bar when the first set of grounding bars are not connected; and

connecting a ground shoe assembly to the lower ground bus bar and configured to engage a grounding circuit carried by the switchgear frame, wherein said ground shoe assembly comprises a ground shoe bracket and first and second opposing bus bars each having first and second ends and carried by said ground shoe bracket, said first ends engaging the grounding circuit, and first and second flexible cables interconnecting each of said second ends to the lower ground bus bar, and die springs carried by said ground shoe bracket and exerting a biasing force against first and second bus bars.

13. The method of claim **12**, wherein said each grounding bar in first and second sets comprises an upper flat end that is configured to engage the respective upper or lower terminal, an outwardly angled section that extends outward

12

from the test device housing, and a vertically extending lower flat section that is configured to engage the lower ground bus bar.

14. The method of claim **12**, wherein the test device housing includes a front panel having removable upper and lower doors to expose upper and lower terminals onto which the grounding bars are secured.

15. The method of claim **12**, wherein the upper and lower terminals comprise upper and lower contact arms each carrying cluster assemblies that engage load and line conductors.

16. The method of claim **12**, wherein each of said first ends of each of said first and second opposing bus bars included flared ends that extend outward, and said grounding circuit includes a grounding bar over which the flared ends engage as the test device housing is inserted within the compartment of the switchgear frame.

17. The method of claim **12**, wherein said first and second flexible cables each comprises a braided cable.

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